

# Railtrack Company Code of Practice

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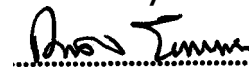
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## METHODOLOGY FOR THE DEMONSTRATION OF ELECTRICAL COMPATIBILITY BETWEEN ROLLING STOCK AND INFRASTRUCTURE

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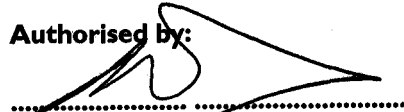
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## ISSUE RECORD

| Issue | Date          | Comments             |
|-------|---------------|----------------------|
| I     | February 2003 | New Code of Practice |

## IMPLEMENTATION

The provisions of this Code of Practice may be used from date of issue.

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## I Purpose

This Code of Practice is intended to assist train-builders, TOCs, and ROSCOs in the production of Route Acceptance Safety Cases for electric trains, by describing a methodology for demonstrating compatibility with EE&CS systems on Railtrack's electrified routes.

## 2 Scope

This Code of Practice is recommended for use by authors of Route Acceptance Safety Cases in accordance with Railway Group Standard GO/RT 3270 Route Acceptance of Rail Vehicles. It is compatible with the guidance and recommendations given in 'Yellow Book 3'.

The methods described in this Code of Practice provide a means of satisfying the requirements of Railway Group Standard GE/RT8015.

This Code of Practice covers Railtrack Controlled Infrastructure routes which are electrified on the 25kV AC overhead system or the 750V DC conductor rail system or which are dual-electrified, and the boundaries to these routes (including non-electrified boundaries).

This Code of Practice covers electrical compatibility from the point of view of interference with the safe operation of EE&CS equipment. It does not cover interference with the satisfactory operation of equipment in cases where there are no safety implications.

This Code of Practice is one of a suite of Codes of Practice (listed in Appendix A) which, when taken together and used in conjunction with a suitable asset register for the routes under consideration, constitute the tools necessary to demonstrate that a particular electric train can safely operate on Railtrack Controlled Infrastructure.

## 3 Definitions & Abbreviations

|               |                                                                                                    |
|---------------|----------------------------------------------------------------------------------------------------|
| EE&CS         | Electrical Engineering & Control Systems                                                           |
| Hazard        | Any situation that could contribute to an accident.                                                |
| ISA           | Independent Safety Assessor                                                                        |
| RASC          | Route Acceptance Safety Case                                                                       |
| RAR           | Railtrack Asset Register                                                                           |
| Route         | The particular section of infrastructure for which acceptance of the rolling stock is being sought |
| Yellow Book 3 | Engineering Safety Management Issue 3                                                              |

## 4 Principle

The 'Methodology' described in this Code of Practice comprises a four-step process, which can be summarised as follows:

- identify the details of the items of EE&CS equipment installed on the route under consideration;
- obtain, from the Codes of Practice within this suite, the relevant electrical susceptibility characteristics for each item or group of items;
- compare the potential interference characteristics of the train with the various limits, or other requirements, contained in the Codes of Practice;
- incorporate the results into a Route Acceptance Safety Case for presentation to the acceptance authority.

These four steps are described in more detail in sections 5.1 - 5.4 below.

## 5 Process

### 5.1 Identification of Equipment on the Route

Details of the EE&CS equipment installed on the route and its boundaries must be obtained. This will typically include:

- types of track circuits;
- types of signalling control and interlocking systems;
- types of remote control systems;
- types of telecomms systems;
- types of lineside equipment including signals, points and lineside circuits;
- type of electrification system.

The preferred method of obtaining this information is for the RASC authors (or their representative) to ask the relevant Railtrack Acceptance Manager for the details of all the types of EE&CS equipment on the required routes and boundaries. Railtrack shall provide the necessary information, either from RAR or from an equivalent source such as Railtrack Record Group Offices.

Initially, it is only necessary to obtain a list of types of equipment installed on the route and boundaries, in order to decide which of the 'Methodology for the Demonstration of Compatibility' documents are relevant to the route.

## 5.2 Obtaining the susceptibility characteristics

For each generic type of equipment which has been identified on the route, the corresponding susceptibility characteristics can be found in the relevant 'Methodology' document, as listed in Appendix A. In each case, the extent to which detailed parameters of the installed equipment (i.e. over and above the simple list of equipment types referred to in 5.1) needs to be obtained can be determined by reference to the 'Methodology' document.

Parameters which are critical to the RASC need to be underwritten by the relevant Zonal Engineer. These parameters shall be identified by the Project, and the Acceptance Manager shall submit them to the Zone Engineer for underwriting.

Each 'Methodology' document gives information on the susceptibility parameters of the equipment concerned; these will generally vary according to the values of the detailed physical parameters of the installed equipment. In some cases, a simple 'pass/fail' benchmark level of tolerable interference is given; in other cases, more complex methodology such as the use of simulation models, or the use of recordings of interference from the train, is prescribed.

## 5.3 Comparison of train interference with susceptibility limits

The electrical interference which the rolling stock can potentially produce must be characterised. This needs to take account of normal steady state and transient operating conditions, and credible degraded mode and fault conditions. Guidance on this specific activity is contained in document RT/E/C /50019 'Guidance on Traction & Auxiliary Monitoring'.

These characteristics are then compared with the susceptibility characteristics contained in the relevant 'Methodology' documents.

Where the train interference can be shown to be within a simple 'pass/fail' limit appropriate to the particular equipment in question in a robust manner, no further demonstration of compatibility is necessary as far as that equipment is concerned.

Where this cannot be shown, or where the nature of the infrastructure equipment is such that no such simple 'pass/fail' criteria can be specified, it will be necessary to carry out further steps, as described in the relevant Methodology document, to produce a satisfactory basis of compatibility. These further steps may require more detailed parameters of the installed equipment to be obtained.

The steps outlined above must be repeated in respect of each of the 'Methodology' documents which have been identified as relevant to the route, so that the compatibility of the rolling stock with all the EE&CS equipment on the route is systematically examined.

The levels in the documents take account of fleet operation. However, further analysis may be required if trains consistently generate interference levels which are near to the limit, and likely to be additive in nature, in respect of those locations (eg major stations and depots) where large numbers of trains are concentrated.

#### **5.4 Incorporation of results into the Route Acceptance Safety Case**

The results of using the 'Methodology' should be incorporated into the RASC by reference to the particular hazards which have been closed out by means of the 'Methodology'.

For example, where a hazard such as 'electrical interference from the train may cause wrong-side failure of DC track circuits' has been closed out by use of the 'Methodology' contained in RT/E/C/50004, this should be explained in the RASC.

## 6 APPENDIX A

### LIST OF METHODOLOGY DOCUMENTS

|              |                                                                                                                                                              |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RT/E/C/50001 | Methodology for the demonstration of electrical compatibility between rolling stock and infrastructure. (Guidance on how to use the compatibility documents) |
| RT/E/C/50002 | Methodology for the demonstration of compatibility with Reed Track Circuits on the AC railway                                                                |
| RT/E/C/50003 | Methodology for the demonstration of compatibility with Reed Track Circuits on the DC railway                                                                |
| RT/E/C/50004 | Methodology for the demonstration of compatibility with DC (AC-immune) Track Circuits                                                                        |
| RT/E/C/50005 | Methodology for the demonstration of compatibility with 50Hz Single Rail Track Circuits                                                                      |
| RT/E/C/50006 | Methodology for the demonstration of compatibility with 50 Hz Double Rail Track Circuits                                                                     |
| RT/E/C/50007 | Methodology for the demonstration of compatibility with HVI Track Circuits                                                                                   |
| RT/E/C/50008 | Methodology for the demonstration of compatibility with TI2I Track Circuits                                                                                  |
| RT/E/C/50009 | Methodology for the demonstration of compatibility with FS2600 Track Circuits                                                                                |
| RT/E/C/50010 | Methodology for the demonstration of compatibility with Non-immune Track Circuits (Aster, DC(NI), Lucas)                                                     |
| RT/E/C/50011 | Methodology for the demonstration of compatibility with Axle Counters                                                                                        |
| RT/E/C/50012 | Methodology for the demonstration of compatibility with TPWS                                                                                                 |
| RT/E/C/50013 | Methodology for the demonstration of compatibility with Interlockings                                                                                        |



|              |                                                                                                  |
|--------------|--------------------------------------------------------------------------------------------------|
| RT/E/C/50014 | Methodology for the demonstration of compatibility with Lineside Equipment on AC and DC Railways |
| RT/E/C/50015 | Methodology for the demonstration of compatibility with Reed FDM Systems                         |
| RT/E/C/50016 | Methodology for the demonstration of compatibility with Telecomms Systems                        |
| RT/E/C/50017 | Methodology for the demonstration of compatibility with Traction Power Supplies & Systems        |
| RT/E/C/50018 | Methodology for the determination of interaction with Neighbouring Railways                      |
| RT/E/C/50019 | Guidance on Traction & Auxiliary Monitoring                                                      |

# RAILTRACK

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## BRIEFING NOTE

**RT/E/C/50001 to RT/E/C/50019**

**Issue 1**

**Methodology for the demonstration of electrical compatibility between rolling stock and infrastructure**

**Issue Date: February 2003**

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### **Background / Synopsis**

This suite of Codes of Practice is intended to assist train-builders, TOCs, and ROSCOs in the production of Route Acceptance Safety Cases for electric trains, by describing a methodology for demonstrating compatibility with EE&CS systems on Railtrack's electrified routes.

The electrical susceptibility of the key items of infrastructure is described and characterised, and recommended methods for demonstrating compatibility between rolling stock and infrastructure are given.

### **Key Changes / Compliance**

This is a new suite of documents. As the documents are Codes of Practice, compliance is not mandatory.

### **Implementation**

The information contained in the documents may be used with immediate effect. The various documents in the suite will be published progressively during 2003.

### **Of Interest To:**

Authors of Route Acceptance Safety Cases.

### **Name and Title of Author**

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